Name:		Index Number:		Class:	

DUNMAN HIGH SCHOOL
Preliminary Examination
Year 6

H2 BIOLOGY — 9648/01

Paper 1 Multiple Choice Paper

22 Sep 2011

1 hour 15 minutes

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST.
Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, civics group and index number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions in this paper. Answer **all** questions. For each question there are four possible answers, **A**, **B**, **C** and **D**.

Choose the **one** you consider correct and record your choice in the **soft pencil** on separate Answer Sheet [**optical answer sheet** (**OTAS**)] provided.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

Calculators may be used.

Do not open this booklet until you are told to do so.

FOR EXAMINER'S USE		
Paper 1		/40
Paper 2	A1	/40
	A2	/40
	B	/20
Paper 3	A1	/20
	A2	/20
	B	/12
	C	/20
Grand Total		/212
Percentage		/%
Grade		

This document consists of **18** printed pages and **0** blank page.

Multiple Choice Questions (40 marks)

Answer **all** questions.

1 The electron micrograph below shows a plant cell.

Which of the following organelle(s) produces adenosine triphosphate?

A D

B C and D

C D and E

D E and F

2 What are the inner folds of the mitochondria called?

A cisternae

B cristae

C granum

D matrix

3 Which of the following is/are the most likely consequence/(s) for a cell lacking functional lysosomes?

(i) The cell becomes crowded with undegraded wastes.
(ii) The cell dies because its ATP-synthesizing mechanisms are missing.
(iii) The cell dies from a lack of enzymes to catalyze metabolic reactions.
(iv) The cell is unable to reproduce itself.
(v) The cell is unable to grow to a mature size and always remains small.

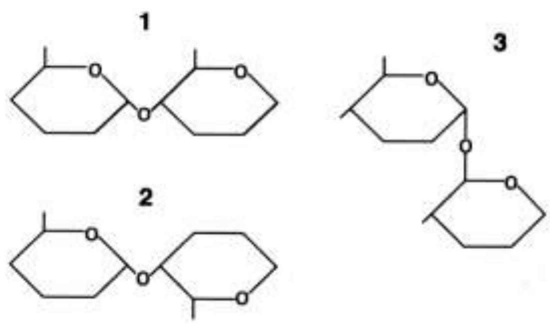
A **(i)** only

B **(i)** and **(v)**

C **(ii)** and **(iv)**

D **(iii)** and **(iv)**

4 The diagrams show different types of bond in polysaccharides.



Which type of bond/(s) is/are found in amylose?

A 1 only

B 2 only

C 1 and 3

D 2 and 3

5 How many different polypeptides, each consisting of r amino acids, can be made if the number of different amino acids available is n?

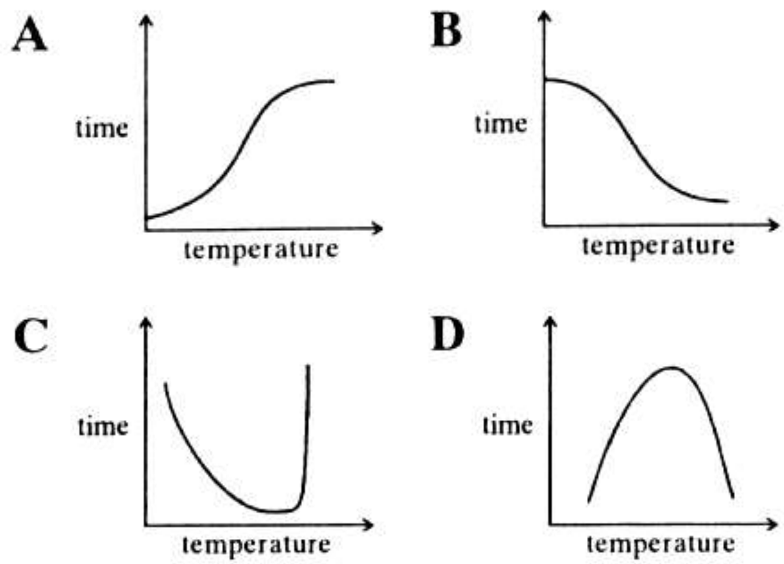
A n^r

B r^n

C nr

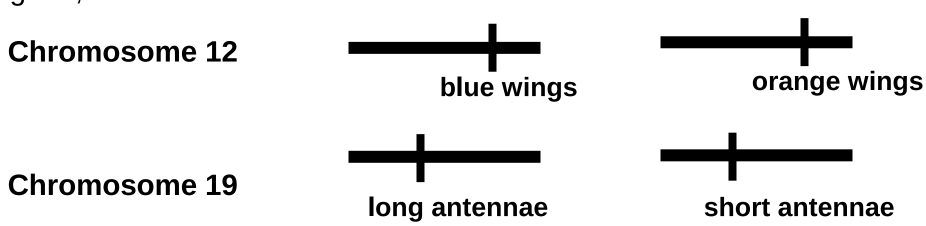
D n

6 In an investigation to determine the effect of temperature on the activity of an enzyme, the time taken for all the substrates to disappear from a standard solution was recorded.



Which graph shows the result of this investigation?

7 A certain (hypothetical) organism is diploid, has either blue or orange wings as the consequence of one of its genes, and has either long or short antennae as the result of a second gene, as shown below.



If an organism is heterozygous for both genes, what will be the possible chromosomes and alleles found in one of her eggs?

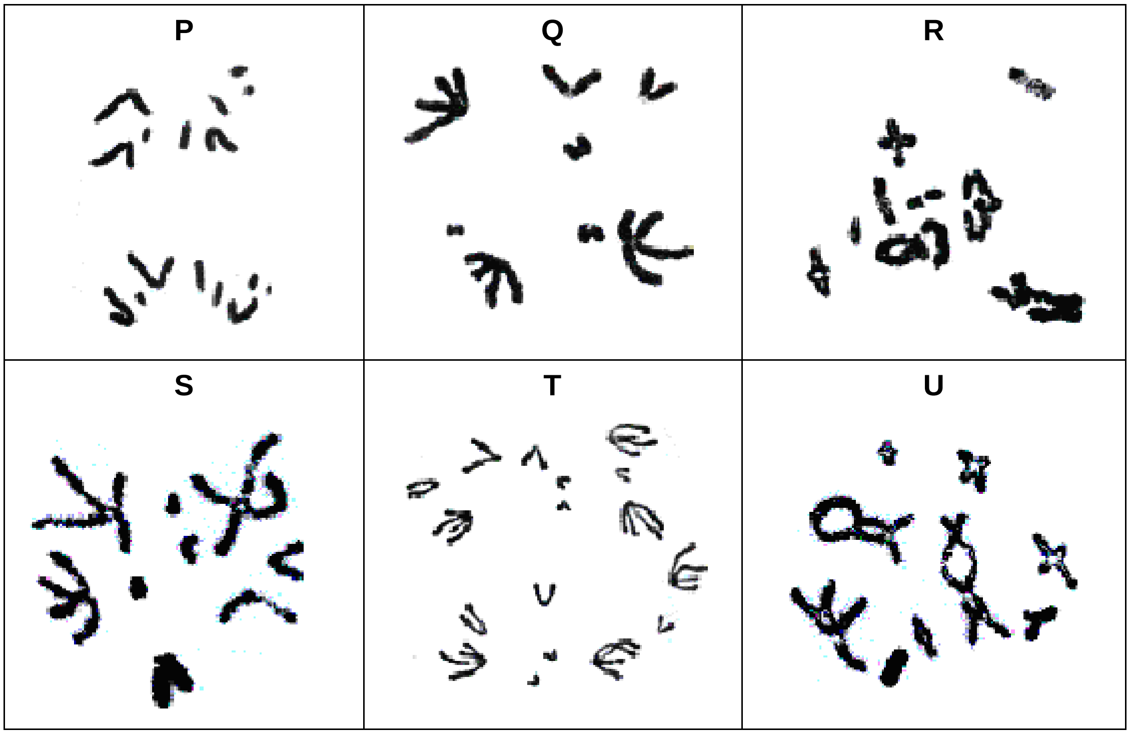
A Two chromosomes 12 with blue and orange wings alleles.

B Chromosome 12 with an orange wings allele and chromosome 19 with long antennae allele.

C Chromosome 19 with both orange wings allele and short antennae allele due to crossing over.

D Chromosome 12 with blue and orange wings alleles and chromosome 19 with long and short antennae alleles.

8 The diagram below shows some of the events that take place during meiosis.



Which of the following shows the correct sequence of events?

A **U → T → R → S → P → Q**

B **U → R → T → S → Q → P**

C **R → Q → U → S → P → T**

D **R → U → Q → S → T → P**

9 Nucleic acids have the following components.

1	**2**	**3**	**4**	**5**
phosphate	ribose	deoxyribose	purine	pyrimidine

Which components are linked by hydrogen bonds in DNA?

A 1 and 3

B 3 and 4

C 2 and 5

D 4 and 5

10 As a ribosome translocates along an mRNA molecule by one codon, which of the following occur(s)?

(i) The tRNA that was in the A-site moves into the P-site.
(ii) The tRNA that was in the P-site moves into the A-site.
(iii) The tRNA that was in the E-site is released.
(iv) A peptide bond is formed between the adjacent amino acids.

A **(i)** and **(ii)**

B **(ii)** and (**iii)**

C **(i)**, **(iii)** and **(iv)**

D **(ii)**, **(iii)** and **(iv)**

11 Two polypeptides were made using synthetic mRNA molecules as shown.

Synthetic mRNA	*Polypeptide produced*
AAAAAAAAAAAA	lysine-lysine-lysine-lysine
UUUAAAUUUAAA	phenylalanine-lysine-phenylalanine-lysine

What are the DNA codes for the amino acids phenylalanine and lysine?

	phenylalanine	**lysine**
A	AAA	TTT
B	AAA	UUU
C	GGG	CCC
D	TTT	GGG

12 Which of the following is present in a bacteriophage but absent in an influenza virus?

A DNA

B RNA

C glycoprotein

D viral capsid

13 Which of the following is/are the way(s) viruses cause diseases in animals?

- **(i)** They inhibit normal host cell DNA, RNA, or protein synthesis.
- **(ii)** They degrade the host cell's chromosomes.
- **(iii)** They disrupt the proto-oncogenes of the host cell causing uncontrolled cell division.
- **(iv)** Their viral proteins and glycoproteins on the surface membrane of host cells cause them to be recognized and destroyed by the body's immune defenses.
- **(v)** They deplete the host cell of cellular materials essential for metabolic functions.

A **(i)**, **(ii)** and **(v)**

B **(i)**, **(ii)**, **(iii)** and **(v)**

C **(i)**, **(ii)**, **(iv)** and **(v)**

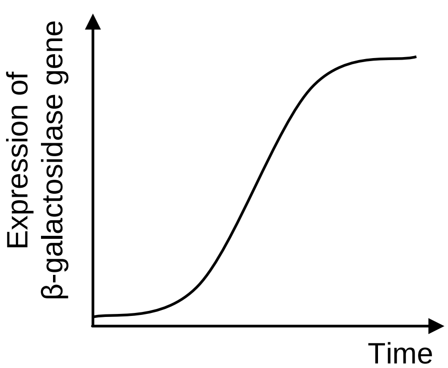
D **(i)**, **(ii)**, **(iii)**, **(iv)** and **(v)**

14 Both lactose and phenyl-β-D-galactose (phenyl-Gal) are substrates for β- galactosidase, but phenyl-Gal does not inactivate the *lac* operon repressor and so is not an inducer.

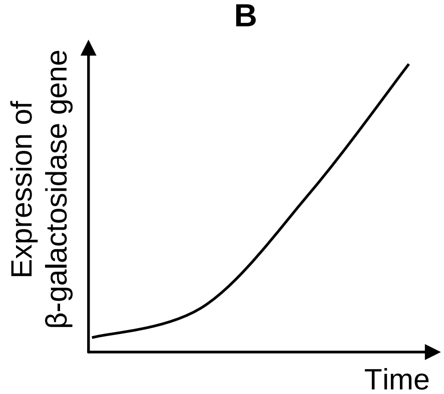
In an experiment, *E. coli* cells were first grown in lactose only for five hours and then in phenyl-Gal only.

Which of the following graphs **BEST** represents the expression profile of β-galactosidase gene over the course of ten hours?

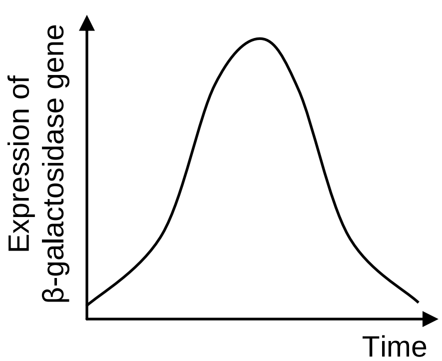
A



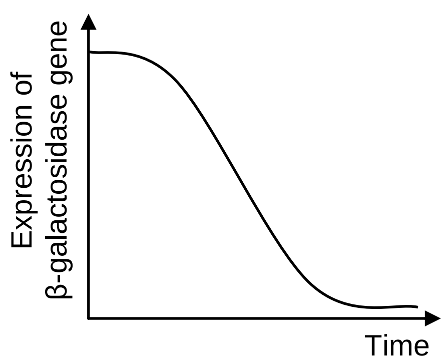
B



C



D



15 Gene amplification is a mechanism known to result in ____________.

(i) synthesis of large amounts of ribosomes
(ii) many copies of the same mRNA coding for drug resistance
(iii) an increased number of gene clusters

Which of the following statements are correct?

A **(i)** and **(iii)**

B **(ii)** and **(iii)**

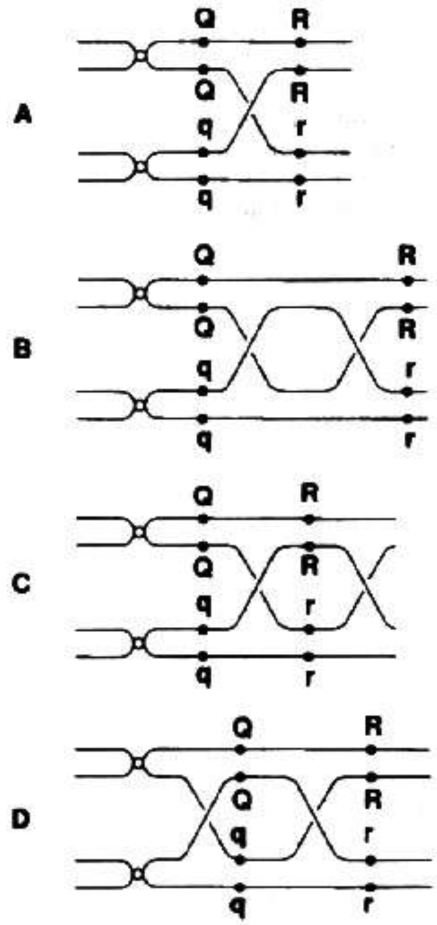
C **(i)** and **(ii)**

D **(i)**, **(ii)** and **(iii)**

16 Which of the following conditions must be present for cancer formation?

	Number of alleles mutated for each proto-oncogene	**Number of alleles mutated for each tumour-suppressor gene**
A	1	1
B	2	1
C	1	2
D	2	2

17 Which diagram represents the situation which would result in the segregation of the alleles **Q** from **q**, and **R** from **r** in the first meiotic division?



18 In cats, black colour is caused by an X-linked allele and the other allele at this locus causes orange colour. The heterozygous is tortoiseshell.

What kinds of offspring would you expect from the cross of a black female and an orange male?

A tortoiseshell female; tortoiseshell male

B black female; orange male

C orange female; orange male

D tortoiseshell female; black male

19 Day-old chicks can be difficult to sex, but their down feathers reveal a sex-linked dominant character called 'barred'. Female birds are **XY** and males are **XX**.

Which cross would ensure correct sexing of the resultant chicks?

	female	**male**
A	barred	homozygous barred
B	barred	homozygous unbarred
C	unbarred	heterozygous barred
D	unbarred	homozygous barred

20 In animals, the amount of energy released from a given quantity of glucose is greater in aerobic respiration than in anaerobic respiration by approximately __________.

A 32 times

B 19 times

C 9 times

D 3 times

21 Which of the following is the role of oxygen in the respiratory electron transfer system?

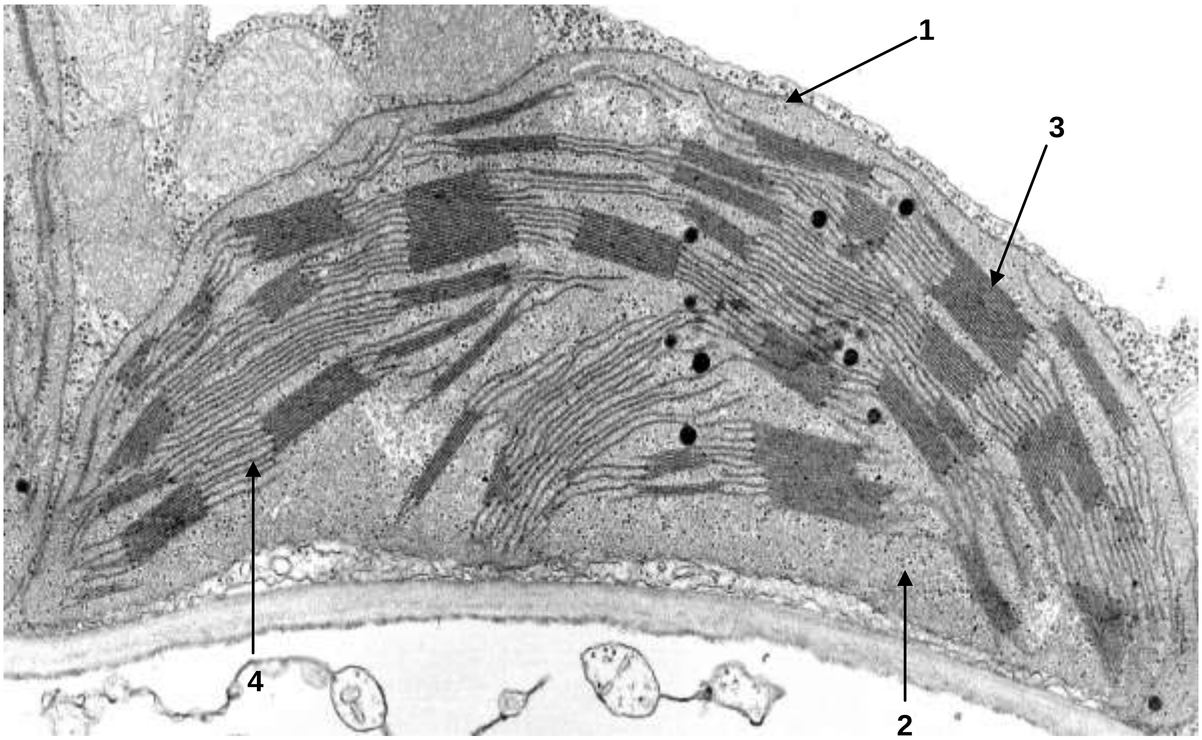
A To accept hydrogen directly from organic acids.

B To accept hydrogen directly from reduced nicotinamide adenine dinucleotide.

C To act as the final acceptor of hydrogen at the end of the chain.

D To donate electrons to the respiratory chain.

22 Which of the following is **CORRECTLY** associated with the function or characteristic of the labelled structure?



	help to maintain high H^+ concentration	made up of phospholipid	has stalked particles embedded in it	where ATP is synthesised
A	1 and 2	1, 2, 3 and 4	4	2
B	3 and 4	2 and 4	2 and 3	1 and 2
C	1 and 4	1, 2 and 3	1 and 4	2 and 4
D	3 and 4	1, 3 and 4	3 and 4	3 and 4

23 The light reaction of photosynthesis does not include __________.

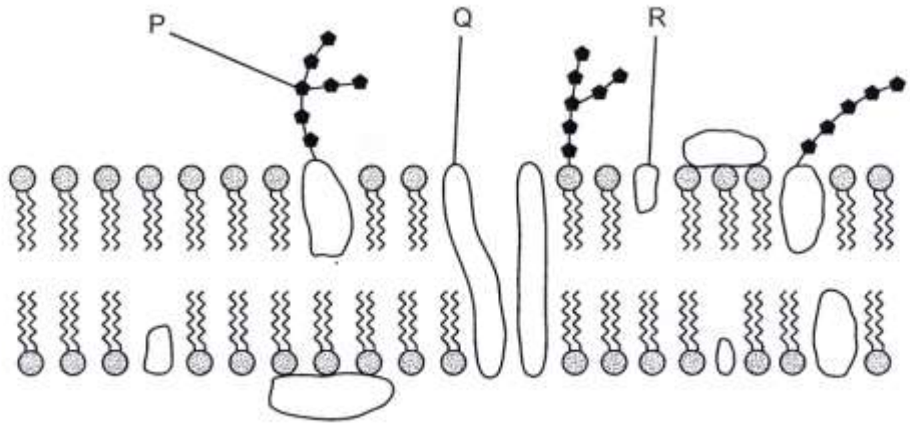
A chemiosmosis

B pigment photoactivation

C NAD reduction

D electron transport

24 The diagram below shows part of a cell surface membrane.



What is the correct function for each of the structures labelled?

	regulates membrane fluidity	**forms hydrogen bonds with water to stabilise membrane**	**transports ions and large polar molecules**
A	R	Q	P
B	P	Q	R
C	Q	R	P
D	R	P	Q

25 Which of the following statements **BEST** describes the outcome of the negative feedback loop in a homeostatic mechanism?

A Glucagon is released by the pancreas in response to a drop in blood glucose concentration below the normal level.

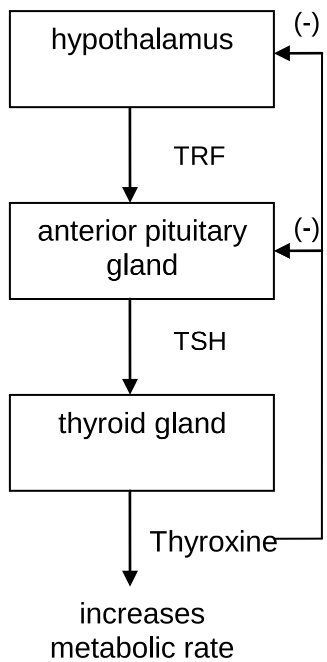
B The liver cells break down glycogen to glucose in response to glucagon.

C Blood glucose level increases to the normal level in response to the release of glucagon.

D Release of glucagon is stopped in response to blood glucose concentration returning to the normal level.

26 Homeostasis often involves multi-step pathways. An example is given in the diagram below, which shows the hormonal feedback mechanism that controls body temperature.

TRF = thyrotrophin-releasing factor; TSH = thyroid-stimulating hormone



A rise in temperature causes a decrease in the amount of TRF released by the hypothalamus. As a result of this, the level of thyroxine will __________.

A rise with a consequent fall in the basic metabolic rate

B fall with a consequent fall in the basic metabolic rate

C inhibit production of TRF

D inhibit production of TSH

27 The vertebrate system conducts impulses only in one direction. This one-way conductance across a synapse occurs __________.

A because the Na^+/K^+ pump moves ions in one direction

B as a result of voltage-gated sodium channels found in the vertebrate system

C because only the postsynaptic membranes can bind neurotransmitters

D because vertebrate nerve cells have dendrites

28 The postsynaptic membrane of a nerve may be stimulated by certain neurotransmitters to permit the influx of negative chloride ions into the cell. This process will result in __________.

A membrane depolarization

B an action potential

C the production of an IPSP

D the production of an EPSP

29 Dinitrophenol is a metabolic poison that prevents the production of ATP. What will happen if a resting axon is treated with such a poison?

A The charge inside the axon membrane becomes more negative.

B The charge outside the axon membrane becomes less positive.

C The charge inside the axon membrane becomes less positive.

D The charge outside the axon membrane does not change.

30 Which of the following statements are **TRUE** about the events involved in glycogen metabolism?

(i) cAMP-activated protein kinase stimulates glycogen synthesis
(ii) kinases are activated by phosphorylation
(iii) cAMP levels are raised during glycogen hydrolysis
(iv) receptors activated by cross-phosphorylation leads to glycogenesis

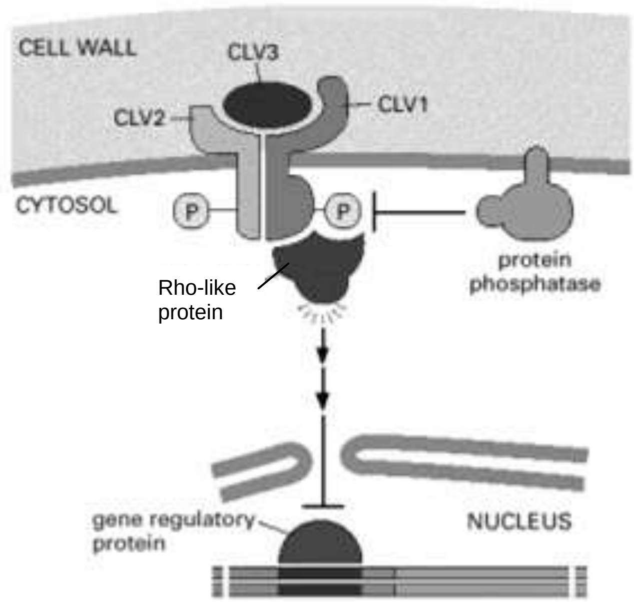
A **(i)** and **(ii)**

B **(ii)** and **(iii)**

C **(i)**, **(iii)** and **(iv)**

D **(ii)**, **(iii)** and **(iv)**

31 The diagram below shows the cell signalling pathway involved in cell proliferation and differentiation in the shoot meristem.



Which of the following statement shows the **CORRECT** sequence of events leading to the activation of the above signalling pathway?

A Dimerisation of CLV2 and CLV1 → binding of CLV3 to dimer → autophosphorylation of serine/threonine residues → activation of Rho-like protein → activation of gene transcription.

B Binding of CLV3 to subunits CLV2 and CLV1→ dimerisation of CLV2 and CLV1 → autophosphorylation of serine/threonine residues → activation of Rho-like protein → activation of gene transcription.

C Binding of CLV3 to subunits CLV2 and CLV1→ dimerisation of CLV2 and CLV1 → dephosphorylation of serine/threonine residues by protein phosphatase → activation of Rho-like protein → activation of gene transcription.

D Dimerisation of CLV2 and CLV1 → binding of CLV3 to dimer → dephosphorylation of serine/threonine residues by protein phosphatase → activation of Rho-like protein → activation of gene transcription.

32 The allele that causes phenylketonuria (PKU) is harmful. What maintains the presence of this harmful allele in a population's gene pool?

A balanced polymorphism

B diploidy in organisms

C heterozygote advantage

D stabilising selection

33 Mutations that occur deep within intronic sequences that are not subject to alternative splicing, and not subject to DNA-repair enzyme activity, should usually be ________ mutations.

A beneficial

B harmful

C neutral

D point

34 The average mutation rate of a gene can be used as a molecular clock. Which type of mutation would cause an underestimate of the time that elapsed since two related species diverged from their common ancestor?

A A mutation to the first base of a nucleotide codon

B A mutation to the third base of a nucleotide codon

C A mutation that returns a mutated nucleotide to its original state

D A mutation that substitutes a different amino acid in place of the original amino acid

35 Which of the following is the **MOST** powerful way of increasing the specificity of a DNA profile analysis?

A Select markers present on the sex chromosomes rather than on the autosomes

B Analyze each marker by PCR rather than RFLP analysis

C Repeat the analysis multiple times

D Increase the number of markers used

[Turn over

36 Which of the following statements describe the purpose of transferring DNA fragments from a gel to a nitrocellulose paper during Southern blotting?

- **(i)** To permanently attach the DNA fragments to a substrate.
- **(ii)** To separate the two complementary DNA strands.
- **(iii)** To transfer only the DNA that is of interest.
- **(iv)** To separate out the PCR products.

A **(i)** only

B **(i)** and **(iii)**

C **(ii)**, **(iii)** and **(iv)**

D **(i)**, **(ii)** and **(iii)**

37 The completion of the Human Genome Project revealed that the human genome contains fewer genes than expected, not so many more than simpler organisms. Which of the following statements could explain this phenomenon?

- **(i)** RNA transcripts of human genes are more likely to undergo alternative splicing
- **(ii)** Polypeptide domains are combined in a variety of ways
- **(iii)** Post-translational processing adds diversity to the resulting polypeptides
- **(iv)** Gene expression patterns in humans are often more complex than those in other organisms

A **(i)** only

B **(i)** and **(ii)**

C **(ii)**, **(iii)** and **(iv)**

D **(i)**, **(ii)**, **(iii)** and **(iv)**

38 Which of the following statement regarding stem cells is **FALSE**?

A One potential side effect of any embryonic stem cell-based therapy is the formation of a tumour.

B Stem cells are present within various organs of the adult body.

C Stem cells can develop into a whole organism when implanted into the womb.

D Stem cells can be grown and expanded indefinitely in culture under appropriate conditions.

39 For which of the following disorders would gene therapy be **LEAST** effective?

A sickle cell anaemia

B Huntington's disease

C cystic fibrosis

D Type II diabetes

40 In terms of containment (prevention of genetic pollution of wild plant or non-GM crop populations), which of the following is an advantage of chloroplast transformation (introducing foreign genes into chloroplasts of host plants) over nuclear transformation?

A Chloroplasts are surrounded by a double membrane.

B There are no chloroplasts in pollen of most plant species.

C Chloroplasts are smaller than the nucleus.

D There is no DNA in chloroplasts.

END OF PAPER

[Turn over